

List of Topics

- Module 1: Distance based ML models and Linear Algebra
 - Linear Algebra Review
 - * Distance Metrics & Norms,
 - * Vectors, Matrices,
 - * Eigenvalues, Eigenvectors and Diagonalization
 - Calculus Review
 - * Derivatives
 - * First and second order equations for optimization (i.e., gradient and Hessian)
 - * Hessian and convexity of functions (i.e., convex optimization)
 - Distance based ML models:
 - * K-Nearest Neighbors (supervised)
 - * K-means (unsupervised)
 - * Decision boundaries, perpendicular bisectors, etc.
- Module 2: Linear Regression
 - Least squares loss function and its properties
 - Overdetermined systems and least squares solution
 - Regularization: Ridge Regression ($\|\cdot\|_2$), Lasso Regression ($\|\cdot\|_1$)
 - Undetermined systems and minimum norm solution
- Module 3: Gradient Descent, Regularization, and Kernel Regression
 - Gradient descent for least squares and implicit regularization
 - Step size and learning rate selection, convergence analysis for gradient descent
 - Regularization: Ridge Regression ($\|\cdot\|_2$), Lasso Regression ($\|\cdot\|_1$)
 - Kernel regression formulation and its properties
 - Kernel trick and representation theorem
 - Solution to the kernel regression problem
- Module 4: Classification
 - Sigmoid Function Properties
 - Logistic Regression
 - Decision Boundaries
 - Loss Function Behavior
 - Least Squares vs Logistic Regression
 - Gradient Analysis
 - Error Metrics: Accuracy, Precision, Recall, F1 Score

Clustering

Problem 1. (K-Means Algorithm Understanding.)

Consider running k-means with $k = 2$ on the following four 1-dimensional points: $\{0, 1, 9, 10\}$.

- a. If we initialize the cluster centers at $\mu_1 = 0$ and $\mu_2 = 1$, what happens in the first iteration? What are the new cluster centers?
- b. What is the final clustering after convergence?
- c. If instead we initialize at $\mu_1 = 0$ and $\mu_2 = 10$, what is the final clustering?
- d. What does this tell you about k-means' dependence on initialization?

Problem 2. (K-Means Objective Function.)

The k-means objective is to minimize:

$$J = \sum_{i=1}^n \|\mathbf{x}_i - \boldsymbol{\mu}_{c_i}\|^2$$

where c_i is the cluster assignment for point $\mathbf{x}_i$ and $\boldsymbol{\mu}_j$ is the center of cluster j .

- a. Show that in the assignment step (fixing centers, updating assignments), assigning each point to its nearest center minimizes J .
- b. Show that in the update step (fixing assignments, updating centers), setting $\boldsymbol{\mu}_j = \frac{1}{|C_j|} \sum_{\mathbf{x}_i \in C_j} \mathbf{x}_i$ minimizes J .
- c. Does the k-means algorithm guarantee finding the global minimum of J ? Why or why not?
- d. Can J ever increase during a k-means iteration?

Linear Regression

Problem 3. (Convexity of the Least Squares Objective.) Consider again

$$J(\theta) = \|X\theta - y\|_2^2.$$

- a. Compute the Hessian matrix $\nabla_{\theta}^2 J(\theta)$.
- b. Prove that $J(\theta)$ is a convex function of θ .
- c. State a condition on X under which $J(\theta)$ is *strictly* convex, and explain what strict convexity implies about the number of minimizers.

Problem 4. (Least Squares Solution.)

For linear regression, we want to find $\mathbf{w}$ that minimizes:

$$J(\theta) = \sum_{i=1}^n (y_i - \theta^\top x_i)^2 = \|\mathbf{y} - X\theta\|^2.$$

- a. Write the derivative $\frac{\partial J}{\partial \mathbf{w}}$ and set it to zero to derive the normal equations.
- b. The solution is $\theta^* = (X^\top X)^{-1} X^\top \mathbf{y}$. Under what condition does this solution not exist?
- c. If two features are perfectly correlated (one is a multiple of the other), what happens to $X^\top X$?
- d. Suppose you have more features than data points ($d > n$). Is $(X^\top X)$ invertible?

Problem 5. (Geometric Interpretation.)

- a. The columns of X span a subspace of $\mathbb{R}^n$. What does the linear regression solution represent geometrically in terms of projection?
- b. What is the relationship between the residual vector $\mathbf{r} = \mathbf{y} - X\theta^*$ and the column space of X ?
- c. If you add a new feature that is a linear combination of existing features, does the prediction $\hat{y} = X\theta^*$ change?
- d. Suppose all your features have mean zero. Does the regression line pass through the origin?

Implicit Regularization & Gradient Descent

Problem 6. (Gradient Descent Finds the Minimum-Norm Solution.) Consider the underdetermined least-squares problem

$$\min_{\theta \in \mathbb{R}^d} \|X\theta - y\|^2, \quad X \in \mathbb{R}^{n \times d}, \ n < d, \ \text{rank}(X) = n.$$

Let gradient descent with constant step size $\eta > 0$ and initialization $\theta_0 = 0$ be given by

$$\theta_{t+1} = \theta_t - \eta X^\top (X\theta_t - y).$$

- Show that every iterate θ_t lies in the row space of X .
- Conclude that the limit point θ_∞ (when it exists) must lie in the row space.
- Explain why the minimum-norm least-squares solution is the *only* least-squares minimizer lying in the row space, and hence must equal θ_∞ .

Problem 7. (Early Stopping Acts Like Ridge.) Let gradient descent on the overdetermined least-squares problem

$$\min_{\theta} \|X\theta - y\|^2.$$

be

$$\theta_{t+1} = \theta_t - \eta X^\top (X\theta_t - y), \quad \theta_0 = 0.$$

Let $A = X^\top X$.

a. Show by induction that

$$\theta_t = \left(I - (I - \eta A)^t \right) A^{-1} X^\top y.$$

(b) Argue that when t is small the factor $I - (I - \eta A)^t$ behaves like $\eta t A$.

(c) Conclude that for small t , gradient descent behaves like the ridge-regularized solution

$$\theta_t \approx \left(A + \frac{1}{\eta t} I \right)^{-1} X^\top y.$$

Explain the trade-off: smaller t corresponds to stronger effective regularization.

Problem 8. (Implicit Bias Toward Max-Margin Separator.) Consider linearly separable data $\{(x_i, y_i)\}_{i=1}^n$ with $y_i \in \{\pm 1\}$. Let logistic regression minimize

$$L(\theta) = \sum_{i=1}^n \log(1 + \exp(-y_i \theta^\top x_i)).$$

Gradient descent is

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t), \quad \theta_0 = 0.$$

Assume the data are separable, so the loss has no finite minimizer.

- a. Show that the gradient can be written as

$$\nabla L(\theta) = - \sum_{i=1}^n \sigma(-y_i \theta^\top x_i) y_i x_i,$$

where $\sigma(z) = 1/(1 + e^{-z})$.

- b. Argue informally that as $\|\theta_t\| \rightarrow \infty$, the gradient is dominated by the “hardest” (smallest-margin) examples.

Note that this suggests that the direction $\theta_t/\|\theta_t\|$ approaches the max-margin separator. Indeed, the gradient descent dynamics focus most on the examples with smallest $y_i \theta^\top x_i / \|\theta\|$, i.e., on the minimum margin. This is the same criterion defining the max-margin separator in SVMs, so the direction of θ_t converges toward that separator.

Problem 9. (Implicit Regularization via Small Step Size.) Let $f(\theta) = \|X\theta - y\|^2$ and gradient descent

$$\theta_{t+1} = \theta_t - \eta \nabla f(\theta_t).$$

- a. Show that the update can be written as

$$\theta_{t+1} - \theta_t = -2\eta X^\top (X\theta_t - y).$$

- b. Suppose η is extremely small. Explain why this prevents the iterates from moving far in directions where $X^\top X$ has large eigenvalues.
- c. Interpret this as an implicit form of regularization that suppresses directions in parameter space in which the loss is highly sensitive.

Regularization & Kernel Regression

Problem 10. (Regularization Intuition.)

Consider adding L2 regularization to logistic regression:

$$\min_{\theta} \sum_{i=1}^n L(\hat{y}_i, y_i) + \lambda \|\theta\|^2$$

where $\lambda > 0$ is the regularization parameter.

- What happens to the optimal weights as $\lambda \rightarrow \infty$?
- What happens to the optimal weights as $\lambda \rightarrow 0$?
- Suppose you have two models with the same training accuracy, but model A has $\|\theta_A\| = 10$ and model B has $\|\theta_B\| = 2$. Which would regularization prefer? Why might this be desirable?
- How does regularization affect the decision boundary's orientation/position when multiple decision boundaries could classify the training data perfectly?

Problem 11. (Effect of L_2 Regularization on the Normal Equations.) Consider ridge regression:

$$\hat{\theta} = \arg \min_{\theta} \|X\theta - y\|^2 + \lambda \|\theta\|^2, \quad \lambda > 0.$$

- a. Show that the first-order optimality condition yields the modified normal equations:

$$(X^\top X + \lambda I)\theta = X^\top y.$$

- b. Prove that the matrix $X^\top X + \lambda I$ is always invertible for any $\lambda > 0$.
- c. Explain why this guarantees uniqueness of the ridge solution even when $X^\top X$ is singular.

Problem 12. (Kernel Ridge Regression Dual Form.) Consider kernel ridge regression with kernel matrix $K \in \mathbb{R}^{n \times n}$:

$$\min_{\alpha \in \mathbb{R}^n} \|K\alpha - y\|^2 + \lambda \alpha^\top K \alpha.$$

- a. Show that the first-order optimality condition yields:

$$(K + \lambda I)\alpha = y.$$

- b. Explain why this means predictions take the form

$$f(x) = k_x^\top \alpha, \quad k_x = (k(x_1, x), \dots, k(x_n, x))^\top.$$

- c. Prove that kernel ridge regression and ridge regression in feature space produce identical predictions.

Problem 13. (Why Kernel Ridge Regression is Smoother for Larger λ .) Let $f_\lambda(x) = k_x^\top (K + \lambda I)^{-1} y$ be the kernel ridge predictor.

- a. Show that as λ increases, the eigenvalues of $(K + \lambda I)^{-1}$ decrease.
- b. Explain why this causes f_λ to become a smoother function (i.e., less sensitive to the labels).
- c. Using eigendecomposition $K = U\Lambda U^\top$, show explicitly that:

$$f_\lambda(x) = \sum_{i=1}^n \frac{u_i^\top y}{\lambda_i + \lambda} u_i^\top k_x.$$

- d. Use the above expression to argue that high-frequency components (small λ_i) are suppressed more heavily.

Problem 1 (Ridge Regression: Solution and Limits). Let $X \in \mathbb{R}^{n \times d}$ and $y \in \mathbb{R}^n$. Consider ridge regression

$$\hat{\theta}_\lambda = \arg \min_{\theta \in \mathbb{R}^d} \left\{ \|X\theta - y\|_2^2 + \lambda \|\theta\|_2^2 \right\}, \quad \lambda > 0.$$

Assume $X^\top X$ is invertible.

(a) Show that

$$\hat{\theta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y.$$

(b) Show that

$$\lim_{\lambda \rightarrow 0^+} \hat{\theta}_\lambda = (X^\top X)^{-1} X^\top y,$$

i.e., as $\lambda \rightarrow 0^+$ we recover the ordinary least-squares solution.

(c) Show that $\hat{\theta}_\lambda \rightarrow 0$ as $\lambda \rightarrow \infty$. Interpret this in terms of the trade-off between data fit and regularization.

Solution. (a) Define

$$f(\theta) = \|X\theta - y\|^2 + \lambda \|\theta\|^2.$$

Expand the gradient:

$$\nabla f(\theta) = 2X^\top(X\theta - y) + 2\lambda\theta = 2(X^\top X + \lambda I)\theta - 2X^\top y.$$

At the minimizer, $\nabla f(\hat{\theta}_\lambda) = 0$, so

$$(X^\top X + \lambda I)\hat{\theta}_\lambda = X^\top y.$$

Since $X^\top X$ is invertible and $\lambda > 0$, the matrix $X^\top X + \lambda I$ is also invertible. Thus

$$\hat{\theta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y.$$

(b) Consider the expression from part (a):

$$\hat{\theta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y.$$

As $\lambda \rightarrow 0^+$, we have $X^\top X + \lambda I \rightarrow X^\top X$ (entrywise), and its inverse converges to $(X^\top X)^{-1}$ because $X^\top X$ is invertible and matrix inversion is continuous at invertible matrices. Hence

$$\lim_{\lambda \rightarrow 0^+} \hat{\theta}_\lambda = \lim_{\lambda \rightarrow 0^+} (X^\top X + \lambda I)^{-1} X^\top y = (X^\top X)^{-1} X^\top y.$$

(c) Again, from part (a),

$$\hat{\theta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y.$$

Factor out λ :

$$X^\top X + \lambda I = \lambda \left(I + \frac{1}{\lambda} X^\top X \right),$$

so

$$(X^\top X + \lambda I)^{-1} = \frac{1}{\lambda} \left(I + \frac{1}{\lambda} X^\top X \right)^{-1}.$$

As $\lambda \rightarrow \infty$, the matrix $\frac{1}{\lambda} X^\top X \rightarrow 0$, so $\left(I + \frac{1}{\lambda} X^\top X \right)^{-1} \rightarrow I$. Therefore

$$(X^\top X + \lambda I)^{-1} \sim \frac{1}{\lambda} I \quad \text{and} \quad \hat{\theta}_\lambda \sim \frac{1}{\lambda} X^\top y \rightarrow 0.$$

Interpretation: as λ grows, we put more weight on the penalty $\|\theta\|^2$ and less on the data fit. The optimizer shrinks toward the zero vector to minimize the size of the parameter.

Problem 14. (Ridge as Least Squares on Augmented Data.) Let $X \in \mathbb{R}^{n \times d}$, $y \in \mathbb{R}^n$, and $\lambda > 0$. Consider the ridge regression objective

$$\min_{\theta \in \mathbb{R}^d} \|X\theta - y\|^2 + \lambda \|\theta\|^2.$$

Define the augmented matrix and vector

$$\tilde{X} = \begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix} \in \mathbb{R}^{(n+d) \times d}, \quad \tilde{y} = \begin{bmatrix} y \\ 0 \end{bmatrix} \in \mathbb{R}^{n+d}.$$

(a) Show that

$$\|X\theta - y\|^2 + \lambda \|\theta\|^2 = \|\tilde{X}\theta - \tilde{y}\|^2.$$

(b) Deduce that ridge regression is equivalent to solving an ordinary least-squares problem on $(\tilde{X}, \tilde{y})$.

Problem 15. (Minimum-Norm Solution Lies in the Span of the Data.) Let $x_1, \dots, x_n \in \mathbb{R}^d$ be data points and $y \in \mathbb{R}^n$. Consider the regularized problem

$$\hat{w} = \arg \min_{w \in \mathbb{R}^d} \left\{ \sum_{i=1}^n (w^\top x_i - y_i)^2 + \lambda \|w\|^2 \right\}, \quad \lambda > 0.$$

Show that the optimal w can be written as a linear combination of the data:

$$\hat{w} = \sum_{i=1}^n \alpha_i x_i \quad \text{for some } \alpha \in \mathbb{R}^n.$$

(Hint: decompose any w into a part in $\text{span}\{x_1, \dots, x_n\}$ and a part orthogonal to that span.)

Problem 16. (Kernel Ridge Regression in the Dual.) Let $\phi : \mathcal{X} \rightarrow \mathbb{R}^m$ be a feature map and $\Phi \in \mathbb{R}^{n \times m}$ the matrix of stacked feature vectors: its i -th row is $\phi(x_i)^\top$. Consider the regularized problem

$$\hat{w} = \arg \min_{w \in \mathbb{R}^m} \{ \|\Phi w - y\|^2 + \lambda \|w\|^2 \}, \quad \lambda > 0.$$

- (a) Show that the optimal w lies in the span of the feature vectors $\phi(x_1), \dots, \phi(x_n)$; i.e., $w = \Phi^\top \alpha$ for some $\alpha \in \mathbb{R}^n$.
- (b) Let $K = \Phi \Phi^\top \in \mathbb{R}^{n \times n}$ with entries $K_{ij} = \phi(x_i)^\top \phi(x_j)$ (the kernel matrix). Show that α satisfies

$$(K + \lambda I) \alpha = y.$$

- (c) Show that the prediction at a new point x_* is

$$\hat{f}(x_*) = \phi(x_*)^\top \hat{w} = k_*^\top \alpha,$$

where $k_* \in \mathbb{R}^n$ has entries $(k_*)_i = \phi(x_i)^\top \phi(x_*)$.

Problem 17. (Centering the Kernel Matrix.) Let $\phi(x) \in \mathbb{R}^m$ be a feature map and $\tilde{\phi}(x) = \phi(x) - \mu$, where

$$\mu = \frac{1}{n} \sum_{i=1}^n \phi(x_i)$$

is the empirical mean of the feature vectors. Define the kernel

$$K_{ij} = \phi(x_i)^\top \phi(x_j),$$

and the centered kernel

$$K_{ij}^c = \tilde{\phi}(x_i)^\top \tilde{\phi}(x_j).$$

Show that in matrix form,

$$K^c = HKH, \quad H = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top,$$

where $\mathbf{1} \in \mathbb{R}^n$ is the all-ones vector.

Least Squares Classification

Problem 18. (Scaling the Labels and Effect on Classification Boundary.) Consider least-squares *classification* with labels $y_i \in \{-1, +1\}$ and linear predictor $f(x) = \theta^\top x$. The classifier is $\hat{y} = \text{sign}(f(x))$.

Now define new labels $\tilde{y}_i = c y_i$ for some constant $c > 0$. Let $\hat{\theta}$ be the least-squares solution using labels y , and let $\tilde{\theta}$ be the least-squares solution using labels $\tilde{y}$.

- a. Show that the normal equations for the two problems are

$$X^\top X \hat{\theta} = X^\top y, \quad X^\top X \tilde{\theta} = X^\top \tilde{y} = c X^\top y.$$

- b. Prove that $\tilde{\theta} = c \hat{\theta}$.

- c. Conclude that the classification boundary (the set $\{x : \theta^\top x = 0\}$) is unchanged when we scale the labels by a positive constant.

Problem 19. (Least Squares with ± 1 Labels Approximates Majority Vote and Margin.) Suppose we use least-squares regression for *classification* with labels $y_i \in \{-1, +1\}$. For this problem, fix a particular feature vector x and consider all training examples whose feature vector is exactly x :

$$\{(x, y_1), \dots, (x, y_k)\}, \quad y_j \in \{-1, +1\}.$$

We fit a scalar prediction $f \in \mathbb{R}$ by minimizing the empirical squared loss

$$L(f) = \sum_{j=1}^k (f - y_j)^2.$$

- a. Show that the minimizer f^* of $L(f)$ is the average of the labels:

$$f^* = \frac{1}{k} \sum_{j=1}^k y_j.$$

- b. Let k_+ be the number of examples with $y_j = +1$ and k_- the number with $y_j = -1$ (so $k_+ + k_- = k$). Show that

$$f^* = \frac{k_+ - k_-}{k} = 2 \frac{k_+}{k} - 1.$$

- c. Conclude that $\text{sign}(f^*)$ agrees with the majority class label among the k examples at x .
- d. Explain why the magnitude $|f^*|$ can be interpreted as a kind of *margin* measuring how strong the majority is.

Problem 20. (Least-Squares Classification: Decision Boundary Invariance.) Let $f_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$ be trained by least squares. Suppose we rescale feature x_2 via $x'_2 = c x_2$ for some constant $c > 0$. Let the resulting parameter vector after training on (x_1, x'_2) be $\theta' = (\theta'_0, \theta'_1, \theta'_2)$.

- a. Prove that the geometric decision boundary in the *original* (x_1, x_2) plane is unchanged by the feature scaling.
- b. Show that $\theta'_2 = \theta_2/c$ must hold at the optimum (no computation needed).
- c. Argue that this implies θ'_0, θ'_1 must equal θ_0, θ_1 .

Problem 21. (Equivalence of Least Squares and Logistic Boundaries.) Consider least-squares classification with predictor

$$f_{\text{LS}}(x) = \theta^\top x,$$

and logistic regression with predictor

$$f_{\text{LR}}(x) = \sigma(\theta^\top x), \quad \sigma(t) = \frac{1}{1 + e^{-t}}.$$

- a. Prove that both models induce the *same* decision boundary in feature space.
- b. Show that the boundary is the hyperplane $\theta^\top x = 0$ for both methods.
- c. Explain why this statement holds even though the models minimize completely different loss functions.

Sigmoid Function Properties & Logistic Regression

Problem 22. (Sigmoid Function Properties.) The logistic sigmoid function is defined as $\sigma(z) = \frac{1}{1+e^{-z}}$.

- a. What is $\sigma(0)$?
- b. What is the relationship between $\sigma(z)$ and $\sigma(-z)$? Prove your answer.
- c. If $z_1 > z_2$, how do $\sigma(z_1)$ and $\sigma(z_2)$ compare?
- d. The derivative of the sigmoid is $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. At what value of z is the sigmoid's slope steepest? What is the maximum slope?

Problem 23. (Decision Boundaries.) Consider a binary classification problem with features $\mathbf{x} = [x_1, x_2]^\top$ and labels $y \in \{0, 1\}$. You train a logistic regression model and obtain weights $\theta = [3, -2]^\top$ and bias $b = 1$.

- a. Write the equation for the decision boundary (where $P(y = 1|\mathbf{x}) = 0.5$).
- b. Sketch the decision boundary in the x_1 - x_2 plane.
- c. For a point $\mathbf{x}^* = [1, 1]^\top$, on which side of the decision boundary does it lie? What class would you predict?
- d. If you multiply all weights by 2 (so $\theta_{new} = [6, -4]^\top$ and $b_{new} = 2$), does the decision boundary change? Explain.

Problem 24. (Loss Function Behavior.) The binary cross-entropy loss for a single example is:

$$L(\hat{y}, y) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

where $y \in \{0, 1\}$ is the true label and $\hat{y} \in (0, 1)$ is the predicted probability.

- a. Suppose the true label is $y = 1$. What is the loss when $\hat{y} = 1$? What about when $\hat{y} \rightarrow 0$?
- b. Suppose the true label is $y = 0$. What is the loss when $\hat{y} = 0$? What about when $\hat{y} \rightarrow 1$?
- c. For $y = 1$, rank the following predictions from lowest to highest loss: $\hat{y} \in \{0.9, 0.6, 0.51\}$.
- d. Explain why this loss function encourages the model to output probabilities close to 0 or 1 for training examples, rather than staying near 0.5.

Problem 25. (Least Squares vs Logistic Regression.)

Consider using least squares for binary classification with labels $y \in \{-1, +1\}$. The least squares objective is:

$$\min_{\theta} \sum_{i=1}^n (\theta^\top \mathbf{x}_i - y_i)^2$$

- a. Suppose you have a training point $(\mathbf{x}_1, y_1 = +1)$ that is correctly classified and very far from the decision boundary, meaning $\theta^\top \mathbf{x}_1 = 10$. What is the contribution of this point to the least squares loss?
- b. If you have another correctly classified point $(\mathbf{x}_2, y_2 = +1)$ closer to the boundary with $\theta^\top \mathbf{x}_2 = 1.5$, which point contributes more to the loss?
- c. Explain why this behavior might be undesirable for classification.
- d. How does logistic regression differ in this regard?

Problem 26. (Gradient Analysis.)

For logistic regression with a single training example $(\mathbf{x}, y)$ where $y \in \{0, 1\}$, the gradient of the loss with respect to the weights is

$$\nabla_{\theta} L = (\hat{y} - y)\mathbf{x}$$

where $\hat{y} = \sigma(\theta^{\top} \mathbf{x} + b)$.

- a. If $y = 1$ and $\hat{y} = 0.9$, what is the sign of the gradient? In which direction will gradient descent move the weights?
- b. If $y = 0$ and $\hat{y} = 0.3$, what is the sign of the gradient? In which direction will gradient descent move the weights?
- c. If $y = 1$ and $\hat{y} = 0.1$ (a bad prediction), how does the magnitude of the gradient compare to part (a)? What does this mean for learning?
- d. What happens to the gradient when $\hat{y} = y$? Interpret this result.

Problem 27. (Multi-class Extension.)

Consider extending binary logistic regression to K classes. One approach is the softmax function, where for class k :

$$P(y = k|\mathbf{x}) = \frac{e^{\theta_k^\top \mathbf{x}}}{\sum_{j=1}^K e^{\theta_j^\top \mathbf{x}}}$$

- a. Show that $\sum_{k=1}^K P(y = k|\mathbf{x}) = 1$ (i.e., the probabilities sum to 1).
- b. For $K = 2$ classes, show that the softmax reduces to binary logistic regression. (Hint: consider the probability ratio.)
- c. If you add the same constant c to all $\mathbf{w}_k^\top \mathbf{x}$ values, do the probabilities change? What does this imply?
- d. If $\mathbf{w}_1^\top \mathbf{x} > \mathbf{w}_2^\top \mathbf{x} > \mathbf{w}_3^\top \mathbf{x}$, which class has the highest probability?

Problem 28. (Sigmoid as the Unique Smooth Monotone Squashing Function.) Let $\sigma(t) = \frac{1}{1+e^{-t}}$. Suppose we want a prediction function $g(t)$ for binary classification satisfying:

- $g(t)$ is smooth and strictly increasing,
 - $\lim_{t \rightarrow -\infty} g(t) = 0$ and $\lim_{t \rightarrow +\infty} g(t) = 1$,
 - the derivative satisfies $g'(t) = g(t)(1 - g(t))$.
- a. Show that the above differential equation uniquely determines $g(t)$ up to horizontal shifting.
 - b. Solve the differential equation to obtain the explicit form of $g(t)$.
 - c. Explain why changing the shift does not change the decision boundary of logistic regression.

Problem 29. (Gradient of Logistic Loss: Derivation and Structure.) Let $\ell(\theta) = -y \log \sigma(\theta^\top x) - (1 - y) \log(1 - \sigma(\theta^\top x))$.

- a. Show that the gradient satisfies

$$\nabla_{\theta} \ell(\theta) = (\sigma(\theta^\top x) - y)x.$$

- b. Argue why this form implies that the gradient always points in the direction of x .
- c. Prove that the logistic loss is minimized when $\sigma(\theta^\top x) = y$ for a single training point.

Error Metrics for Classification

Problem 30. (Confusion Matrix and Interpretation.) You observe the following outcomes of a classifier (order does not matter):

TP, TN, FP, FN, TN, TP.

1. Construct the 2×2 confusion matrix (no counts needed beyond tallies).
2. State which type of error (FP or FN) represents predicting $+1$ when the truth is -1 .
3. Explain in words the difference between precision and recall, without using probability.

Problem 31. (Precision and Recall Under Threshold Changes.)

A binary classifier assigns each point a score $s(x) \in \mathbb{R}$, and predicts $\hat{y} = +1$ when $s(x) \geq t$ and -1 otherwise. Consider what happens when you **increase** the threshold t .

- a. Without using probability, argue which metric must (weakly) increase: precision or recall.
- b. Explain intuitively why the other metric must (weakly) decrease.

Problem 32. (Interpreting the F1 Score.)

The F1 score of a classifier is defined as

$$\text{F1} = \frac{2}{\frac{1}{\text{precision}} + \frac{1}{\text{recall}}}.$$

- a. Show algebraically that F1 is the harmonic mean of precision and recall.
- b. Explain why the F1 score is small when either precision or recall is small.

Problem 33. (Accuracy Can Be Misleading.)

A dataset has 990 points with label -1 and 10 points with label $+1$. A classifier always outputs $\hat{y} = -1$.

- a. Compute the accuracy.
- b. Compute precision and recall for the $+1$ class.
- c. Explain why accuracy gives a misleading impression of performance.

Problem 34. (ROC Curve: Why FPR Must Increase Before TPR.)

For a fixed scoring function $s(x)$, consider thresholds $t_1 < t_2$ where the classifier predicts +1 when $s(x) \geq t$.

- a. Show that lowering the threshold from t_2 to t_1 cannot decrease the true positive rate (TPR).
- b. Show that lowering the threshold cannot decrease the false positive rate (FPR) either.
- c. Interpret these facts in terms of the ROC curve always moving “up and to the right” as threshold decreases.

Problem 35. (Effect of Class Imbalance on Metrics.)

A classifier predicts $\hat{y} = +1$ for 5 points and $\hat{y} = -1$ for the remaining 95 points. Among the 5 predicted positives, 3 are true positives. In total, there are 20 actual +1 points.

- a. Construct the confusion matrix.
- b. Compute precision and recall.
- c. The classifier is correct on 93 points. Compute accuracy and explain why it hides the model's poor performance.